# Week 2 Update

Michael Fang

March 2026



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Weekly progress meeting with Professor Torquato.
Main things to cover:

- What I finished from last week
- New patterns I analyzed (Classes II, III)
- The Bombieri-Taylor result
- Updated ranking

# Quick Recap: What I Had Last Week

Last week I had:

- Variance code validated (Poisson benchmark, 1.1% error)
- Three metallic-mean chains: Fibonacci, Silver, Bronze
- All gave $\alpha = 3$ as expected

| **Pattern** | $\alpha$ (measured) | $\alpha$ (expected) | $\bar{\Lambda}$ |
|---|---|---|---|
| Fibonacci | 3.05 | 3 | 0.200 |
| Silver | 2.99 | 3 | 0.250 |
| Bronze | 2.99 | 3 | 0.282 |

**This week:** I wanted to cover all three hyperuniformity classes and find something in the $1 < \alpha < 2$ gap.





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**This week:** I wanted to cover all three hyperuniformity classes and find something in the $1 < \alpha < 2$ gap.

Quick reminder of where we left off.
The metallic means all give $\alpha = 3$ because of the 2x2 matrix constraint: $|\lambda_2| = 1/|\lambda_1|$.
Goal this week was to expand beyond just $\alpha = 3$.

# Class II: Period-Doubling

I implemented the Period-Doubling chain:

$$S \to L, \quad L \to LSS \qquad \Rightarrow \qquad M = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$$

Matrix gives $\lambda_1 = 2$, $|\lambda_2| = 1$

So: $\alpha = 1 - \dfrac{2\ln(1)}{\ln(2)} = 1$

**What I found:**

- Variance grows like $\ln R$ (not bounded, not power-law)
- $\bar{\Lambda}$ diverges — can't use it to rank this pattern
- This is the boundary case: $\alpha = 1$ exactly
- Class II confirmed





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Period-Doubling is the boundary case: $\alpha = 1$ exactly.
$|\lambda_2| = 1$ means $\ln(1) = 0$ in the formula.
This is our first Class II example — variance grows logarithmically.

# Class III: 0222 Chain

Also implemented the 0222 chain:

$$S \to LL, \quad L \to SSLL$$

Here $|\lambda_2| = \sqrt{5} - 1 > 1$
So: $\alpha \approx 0.64$

**What I found:**

- Variance grows like $R^{0.36}$
- $\bar{\Lambda}$ diverges
- Class III confirmed

Now I have examples of all three classes.

**The three classes:**

| Class | $\alpha$ | $\sigma^2(R)$ |
|---|---|---|
| I | $> 1$ | bounded |
| II | $= 1$ | $\sim \ln R$ |
| III | $< 1$ | $\sim R^{1-\alpha}$ |

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0222 has $|\lambda_2| > 1$, which gives $\alpha < 1$.
The power-law growth $R^{0.36}$ matches the prediction $R^{1-\alpha}$.
Now we have all three classes covered.

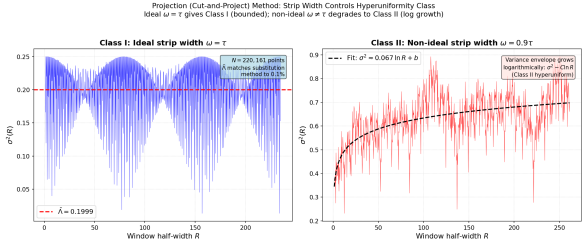
# URL Verification

I also tested the perturbed lattice formula you mentioned:

$$\bar{\Lambda}(a) = \frac{1}{6} + \frac{a^2}{3}$$

**My numerical results:**

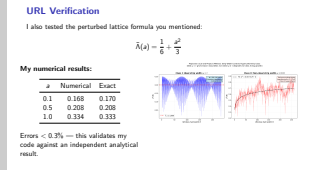
| $a$ | Numerical | Exact |
|---|---|---|
| 0.1 | 0.168 | 0.170 |
| 0.5 | 0.208 | 0.208 |
| 1.0 | 0.334 | 0.333 |




Errors $< 0.3\%$ — this validates my code against an independent analytical result.







URL = uncorrelated random lattice (independent displacements).
Good validation because we have the exact formula.
Always $\alpha = 2$ regardless of displacement amplitude $a$.

# Stealthy Patterns

I analyzed stealthy configurations provided by another group member.

**Idea:** Force $S(k) = 0$ for all $|k| < K^*$, not just at $k = 0$.

**Results** ($N = 2{,}000$, $\sim$4000 configs each):

| $\chi$ | $\bar{\Lambda}$ | $\alpha$ |
|---|---|---|
| 0.1 | 0.166 | 2.0 |
| 0.2 | 0.166 | 2.0 |
| 0.3 | 0.166 | 2.0 |

**Key observation:** $\bar{\Lambda} \approx 1/6$ — same as the perfect lattice!

These disordered patterns achieve optimal hyperuniformity.





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Data from another grad student (CCO format, 4000 configs per $\chi$).
$\chi$ controls the size of the exclusion region.
Remarkable: disordered patterns achieve lattice-level $\bar{\Lambda}$.

# The Gap Problem

At this point I noticed a gap in my data:

| $\alpha$ | Class | What I have |
|---|---|---|
| 3 | I | Fibonacci, Silver, Bronze |
| 2 | I | URL, Stealthy |
| $1 < \alpha < 2$ | I | Nothing! |
| 1 | II | Period-Doubling |
| $< 1$ | III | 0222 |

The problem: all $2\times2$ metallic-mean matrices give $\alpha = 3$ because $|\lambda_2| = 1/|\lambda_1|$.

You suggested looking at the Bombieri-Taylor cubic substitution from their 1986 paper.





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The problem: all $2\times2$ metallic-mean matrices give $\alpha = 3$ because $|\lambda_2| = 1/|\lambda_1|$.
You suggested looking at the Bombieri-Taylor cubic substitution from their 1986 paper.

The 2x2 constraint: $\det(M) = \pm1$ forces $|\lambda_2| = 1/|\lambda_1|$.
This always gives $\alpha = 1 + 2 = 3$.
Need a 3x3 matrix to break this constraint.

# Bombieri-Taylor Cubic Substitution

From the 1986 paper (p. C3-21):

**Three-letter alphabet:**

$$a \to aac,\ b \to ac,\ c \to b$$

**Matrix:**

$$M = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Eigenvalues: $\theta_1 = 2.247$, $|\lambda_2| = 0.802$

**Predicted:**

$$\alpha = 1 - \frac{2\ln(0.802)}{\ln(2.247)} = \mathbf{1.545}$$

**What I did:**

- Extended my code for 3-letter alphabets
- Generated 9.5 million points
- Computed variance and $\bar{\Lambda}$

**Result:** $\bar{\Lambda} = 0.377$, Class I confirmed.





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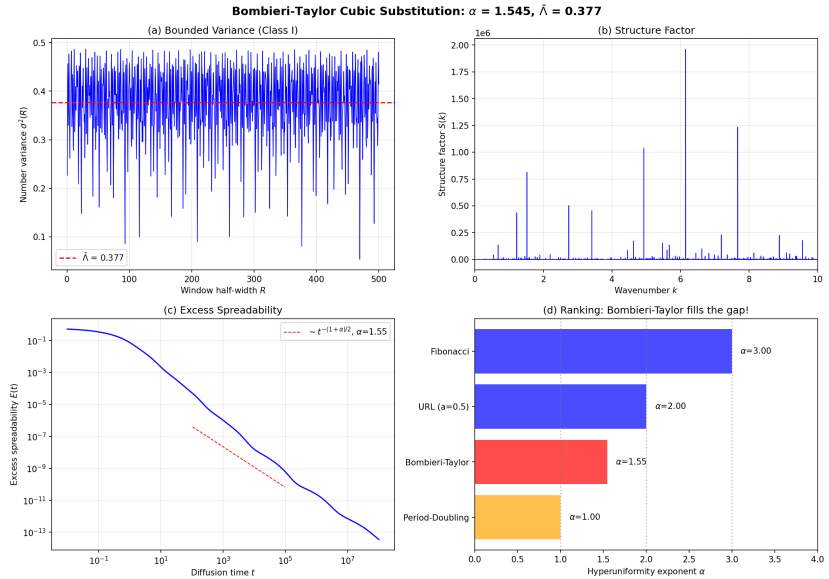
Result: $\bar{\Lambda} = 0.377$, Class I confirmed.

Key: $|\lambda_2| = 0.802 \neq 1/\theta_1 = 0.445$.
The 3x3 matrix breaks the constraint!
$\theta_1$ is a Pisot number (conjugates all $< 1$).

# Bombieri-Taylor Results




Variance is bounded (Class I). The log-periodic oscillations in spreadability confirm the

Panel (a): Bounded variance confirms Class I.
Panel (c): Log-periodic oscillations with period $\ln(\theta_1)$.
Panel (d): Shows where B-T fits in the ranking.
The eigenvalue formula $\alpha = 1.545$ is exact.

# Updated Ranking

Here's where everything stands now:

| Pattern | Class | $\alpha$ | $\bar{\Lambda}$ |
|---|---|---|---|
| Lattice | I | $\infty$ | 0.167 |
| Stealthy | I | 2.0 | 0.166 |
| URL ($a = 0.1$) | I | 2.0 | 0.168 |
| Fibonacci | I | 3.0 | 0.200 |
| Silver | I | 3.0 | 0.250 |
| Bronze | I | 3.0 | 0.282 |
| URL ($a = 1.0$) | I | 2.0 | 0.333 |
| **Bombieri-Taylor** | **I** | **1.545** | **0.377** |
| Period-Doubling | II | 1.0 | diverges |
| 0222 | III | 0.64 | diverges |





Updated Ranking
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URL ($a = 1.0$) | I | 2.0 | 0.333
Bombieri-Taylor | I | 1.545 | 0.377
Period-Doubling | II | 1.0 | diverges
0222 | III | 0.64 | diverges

26 patterns total now (counting different $\chi$ and $a$ values).
Bombieri-Taylor fills the gap between $\alpha = 1$ and $\alpha = 2$.
This is the first quasicrystal I know of with $1 < \alpha < 2$.

# Summary

**What I did this week:**

- Added Class II (Period-Doubling, $\alpha = 1$)
- Added Class III (0222, $\alpha = 0.64$)
- Verified URL formula to $< 0.3\%$
- Analyzed stealthy data from group ($\bar{\Lambda} \approx 1/6$)
- Implemented Bombieri-Taylor cubic ($\alpha = 1.545$)

**Main result:** Filled the $1 < \alpha < 2$ gap with a quasicrystal.

**Questions for you:**

- Are there other cubic/quartic substitutions I should try?
- Should I look at 2D generalizations next?





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Main result: Filled the $1 < \alpha < 2$ gap with a quasicrystal.

Questions for you:
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- Should I look at 2D generalizations next?

Total patterns analyzed: 26
Key accomplishment: Bombieri-Taylor fills the gap.
Possible next steps: other cubic substitutions, 2D/3D.

# References


- Torquato & Stillinger (2003), Phys. Rev. E 68, 041113
- Bombieri & Taylor (1986), J. Physique Colloques 47, C3-19
- Oğuz et al. (2017), Phys. Rev. B 95, 054119
- Torquato (2021), Phys. Rev. E 104, 054102






References

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Main references used this week.